Dashboard / Start Here

Where to begin, how to print, and how to track progress.

Final Chapter 3 update added: transcript-specific teaching points, HP calculator instructions, revision question worked answers, the Slide 57 shape diagram, the variance proof, active-recall study advice, and objective/ethical reporting reminders.

Final Boss Hub

SDS188 A2S1 Chapter 3 Final Boss Study Hub

Numerical Descriptive Measures + Shape + Association

Last updated: 30 May 2026 — final transcript, calculator, revision questions, diagram + variance proof update

How to use this hub

1. Learn what each measure describes: centre, spread, position, shape, grouped data, or relationship.
2. Work through the formula decoder before touching the MCQs.
3. Use the question cards to learn the calculation method + common traps.
4. Complete the mini MCQ set and log every mistake.
5. Use the final checklist in the last 45 minutes before the test.

45-minute pre-test workflow

1. **10 min:** Formula decoder + recognition table.
2. **10 min:** Variance vs SD, CV, z-scores, quartiles/IQR.
3. **10 min:** Grouped data + covariance/correlation cards.
4. **10 min:** Mixed drill + danger words.
5. **5 min:** Error log top weaknesses + one-page checklist.

Depth layer added

This version adds deeper explanation to the high-yield Chapter 3 concepts that are most likely to cause MCQ mistakes: variance vs SD, CV, z-scores, quartiles/IQR, grouped data, empirical vs Chebyshev, covariance/correlation, and transformations.

Note: checked “Mark as understood” boxes are stored in your browser’s localStorage, not inside the HTML file itself. So this update deepens the important concepts rather than trying to read your private local checkbox state.

Progress tracker

Formula Decoder	Measure Recognition Table	Central Tendency Cards
Variation Cards	Z-Scores + Quartiles + Boxplots	Grouped Data Cards
Empirical + Chebyshev Cards	Covariance + Correlation Cards	Transformation Cards
Mixed Drill	Trap Sheet	Mini MCQ Set
Error Log	Final Transcript Additions	

Final Transcript + Revision Additions

Everything added from the final Chapter 3 transcript pack, calculator sheet, revision memo, diagram, and variance proof.

What was added in this final update

This section is a **final enrichment layer**. It does not replace the original notes; it adds the high-yield ideas that were missing or not detailed enough in the previous Chapter 3 HTML.

Calculator layer

HP steps for one-variable data, two-variable data, summaries, correlation, and covariance workaround.

Revision examples

Worked answers from the Chapter 3 revision memo: shares, grouped data, z-scores, stem-and-leaf, cereals, Chebyshev, covariance.

Proof layer

Why the computational variance formula equals the definition formula, without making it exam-heavy.

Interpretation layer

Objective reporting, correlation ≠ causation, active recall, and avoiding passive "memo next to question" studying.

HP Calculator Chapter 3 Workflow

This is for speed-checking Chapter 3 calculations. You still need to understand the formula logic for A2, but the calculator helps verify arithmetic.

Task	HP-style steps / output	Exam trap
Set 4 decimals	Orange downshift → **= / DISP** → **4**.	Most answers are expected to 4 decimals.
Clear memory	Orange downshift → **C / C ALL**.	Do not clear if you still need stored data.
Enter negative number	Type the number first, then press **+/-**.	Do not use the subtraction key for a negative sign.
One-variable data entry	Enter each x-value → **Σ+**. The screen count should increase.	Check n before calculating anything.
One-variable example	X: -6.1, -2.8, 4.3, 7.6, 12.9 → n=5, Σx=15.9, Σx^2=287.71, $\bar{x}$=3.18, s=7.6998.	Use sample SD, not population SD.
Two-variable data entry	Enter x → **INPUT** → enter y → **Σ+**. Repeat for each pair.	Keep x,y pairs together.
Two-variable example	X: 1.2, 1.8, 1.25, 1.72, 1.84, 1.99; Y: 50, 75.2, 42, 72.5, 90.3, 70. Outputs: n=6, Σx=9.8, Σy=400, $\bar{x}$=1.6333, $\bar{y}$=66.6667, s_x=0.3286, s_y=17.6789.	Use SWAP when a button displays both x and y results.
Correlation	Use the calculator correlation function. For the two-variable example: r = 0.8496.	r must be between -1 and 1.
Covariance workaround	No direct covariance button: use `cov(x,y) = r × sₓ × sᵧ`. For the example: 0.8496 × 0.3286 × 17.6789 = 4.9356.	Do not interpret covariance strength directly.

Calculator sanity checks

- If r is outside [-1,1], something has gone wrong.
- If n is not the number of observations/pairs you expected, re-enter the data.

- A comma on the HP display may be a thousands separator; the full stop is the decimal separator.

Variance Proof: Why the Shortcut Formula Works

The proof is not something to memorise as an exam-heavy section, but it helps you trust the shortcut formula and understand why both variance formulas produce the same answer.

Definition formula: $s^2 = [\Sigma(x_i - \bar{x})^2] / (n - 1)$

Computational formula: $s^2 = \{\Sigma x_i^2 - [(\Sigma x_i)^2 / n]\} / (n - 1)$

Deeper explanation

The proof idea in plain English

The definition formula directly measures each value's squared distance from the mean. The computational formula reaches the same result using only Σx_i^2, Σx_i, and n.

1. Start with $\Sigma(x_i - \bar{x})^2$.
2. Expand the squared bracket: $(x_i - \bar{x})^2 = x_i^2 - 2\bar{x}x_i + \bar{x}^2$.
3. Apply the summation to each part: $\Sigma x_i^2 - 2\bar{x}\Sigma x_i + \Sigma\bar{x}^2$.
4. Since $\bar{x}$ is fixed once the sample is known, treat it as a constant.
5. Use $\Sigma x_i = n\bar{x}$ and $\Sigma\bar{x}^2 = n\bar{x}^2$.
6. The middle terms simplify to $\Sigma x_i^2 - n\bar{x}^2$.
7. Rewrite $\bar{x}$ as $(\Sigma x_i)/n$, giving $\Sigma x_i^2 - (\Sigma x_i)^2/n$.

Definition formula explains the meaning. Computational formula is faster for calculation.

Revision Booster 1: Share L — Mean, Geometric Mean, Variance, SD + CV

Data: share price L over 8 time periods: 1720, 1880, 2780, 2040, 2060, 2860, 2230, 1370.

Question	Answer	Why it matters
Mean share price	$\bar{x} = 2117.5$	Ordinary arithmetic average.
Geometric mean share price	$\bar{x}_G = 2064.56$	Multiplicative average of the price levels.
Geometric mean rate of return	$\bar{r}_G = -0.028063 \approx -2.8063\%$	Average compound growth rate across years.
Sample variance	$s^2 = 255050$	Squared spread of share prices.
Standard deviation	$s = 505.0248$	Original-unit spread in share price units.
CV	**23.85%**	Relative variation: s divided by mean.

Trap check: CV uses *standard deviation*, not variance. Geometric return uses growth factors, not raw percentage averaging.

Revision Booster 2: Grouped Data Commodity U

This is the full revision version of grouped mean and variance. It strengthens the smaller grouped-data example already in the hub.

Price range	f	midpoint x	f·x	x^2	$f \cdot x^2$
1-20	2	10.5	21	110.25	220.5
21-40	6	30.5	183	930.25	5581.5
41-60	7	50.5	353.5	2550.25	17851.75
61-80	8	70.5	564	4970.25	39762
81-100	4	90.5	362	8190.25	32761
101-120	3	110.5	331.5	12210.25	36630.75
Total	**n=30**	**-**	**1815**	**-**	**132807.5**

$$\bar{x} \approx 1815 / 30 = 60.5$$

$$s^2 \approx [132807.5 - (1/30)(1815)^2] / 29 = 793.1034$$

$$s \approx \sqrt{793.1034} = 28.1621$$

Deeper explanation

Why the answer is approximate

You only know how many values fall in each price range. You do not know the exact raw prices, so each class midpoint acts as the representative value.

Why the mean is 60.5

The highest frequencies are in the 41-60 and 61-80 classes. The approximate mean falls between those high-frequency classes, which is a good logic check.

Why the SD is large

The standard deviation is about 28.16, nearly half the mean of 60.5. This tells you the prices are fairly spread out, especially because there is frequency toward the edges of the grouped distribution.

> **Grouped data = midpoint × frequency, and every answer is approximate.**

Revision Booster 3: Z-Scores, Stem-and-Leaf + Five-Number Summary

Z-score outlier example

Given X with mean 50 and variance 25, the standard deviation is 5.

x	z = (x−50)/5	Interpretation
27.5	-4.5	Outlier/unusual because $\|z\| > 3$.
49	-0.2	Very close to the mean.
60	2	Above mean, but not beyond 3 SDs.

Stem-and-leaf example

From the revision stem-and-leaf diagram:

Minimum	13
Maximum	69
Range	56
Median	29.5
Mode	29
Shape	Right-skewed

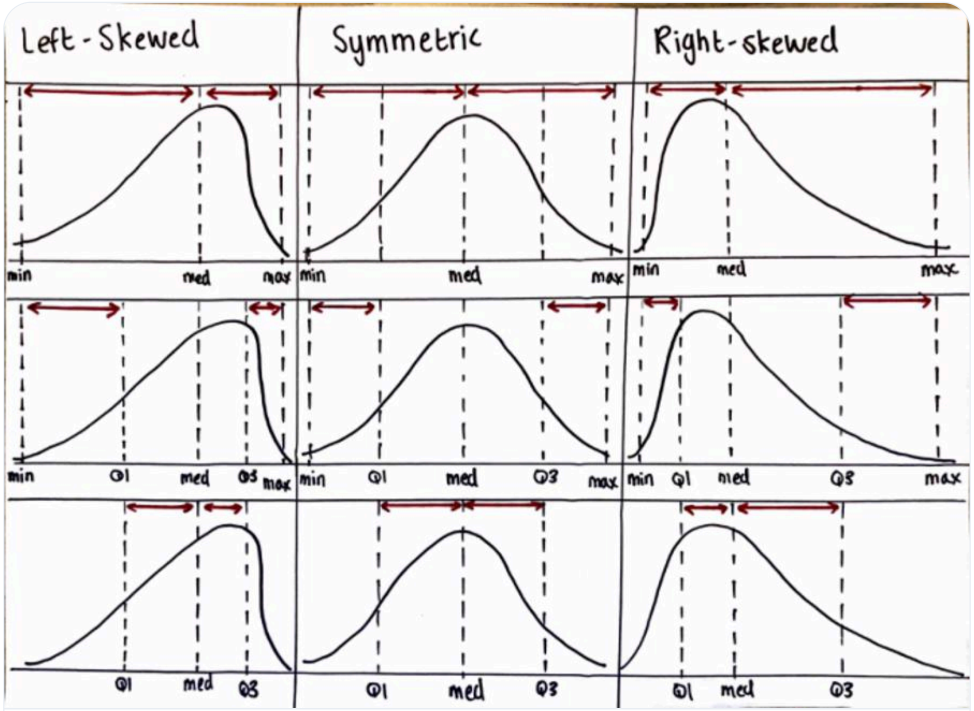
Breakfast cereal five-number summary

> **Minimum = 80, Q_1 = 100, Q_2/Median = 110, Q_3 = 190, Maximum = 200**

> **Trap check:** for z-scores, variance 25 means SD = 5. For stem-and-leaf, reconstruct/order the data before finding median and mode.

Slide 57 Shape Diagram Decoder

The visual point of the diagram is that skewness follows the **tail**, and the five-number summary positions reveal how the data is spread.



Slide 57 visual decoder: left-skewed has a long left tail, symmetric is balanced, right-skewed has a long right tail. The marked positions show how min, Q1, median, Q3 and max shift with shape.

Shape	Visual clue	Typical interpretation
Left-skewed	Long tail to the left; lower side stretched out.	Mean is often pulled below the median.
Symmetric	Balanced left and right sides.	Mean and median are usually close.
Right-skewed	Long tail to the right; upper side stretched out.	Mean is often pulled above the median.

Skewness follows the tail, not the pile of data.

Revision Booster 4: Empirical Rule vs Chebyshev

SAT empirical-rule example

Mean = 500, SD = 90, and the distribution is bell-shaped.

Interval	Bounds	Approx. proportion
$\mu \pm 1\sigma$	410 to 590	68%
$\mu \pm 2\sigma$	320 to 680	95%
$\mu \pm 3\sigma$	230 to 770	99.7%

Cola bottle comparison

Mean fill-weight = 2.06 litres, SD = 0.02 litres.

Rule	Allowed statement
Empirical rule if bell-shaped	About 99.7% are between 2.00 and 2.12 litres, so less than 2 litres is highly unlikely.
Chebyshev if shape unknown	At least 88.89% are between 2.00 and 2.12 litres, so you can only say between 0% and 11.11% may be below 2 litres.

Chebyshev 95% bounds when shape is unknown

```
1 − 1/k² = 0.95 → k² = 20 → k = √20 ≈ 4.47
```

```
Mean = 50, SD = 5: 50 − 4.47(5) = 27.65 50 + 4.47(5) = 72.35
```

Final statement: at least 95% of the data lies between 27.65 and 72.35.

Trap check: Empirical rule is specific but needs bell-shaped data. Chebyshev works for any shape but only gives an *at least* minimum.

Revision Booster 5: Covariance + Correlation Hand Calculation

Revision data: X = 16, 18, 18, 19, 21, 14, 18, 26, 17, 21 and Y = 47, 48, 57, 50, 50, 58, 52, 43, 46, 47.

$$\Sigma x = 188,\ \Sigma y = 498,\ \Sigma xy = 9277,\ n = 10$$

$$\text{cov}(x,y) = [\Sigma xy - (1/n)(\Sigma x)(\Sigma y)] / (n-1)$$

$$\text{cov}(x,y) = [9277 - (1/10)(188)(498)] / 9 = -9.4889$$

$$r = \text{cov}(x,y)/(s_x s_y) = -9.4889 / (3.2931 \times 4.7563) = -0.6058$$

Measure	Interpretation
Covariance = -9.4889	Negative linear relationship: as one variable increases, the other tends to decrease.
Disadvantage of covariance	It shows direction, but not relative strength, because it is unbounded and unit-dependent.
Correlation = -0.6058	Moderately strong negative linear relationship.

Trap check: keep x,y pairs together. Do not sort columns separately. Correlation measures linear relationship; it does not prove causation.

Active Recall + Ethical Reporting Notes

Study method warning

The lecturer specifically warned against sitting with the question and memo side-by-side and thinking, “I would have done that.” That creates false confidence.

1. Attempt the question first without the memo.
2. Mark it afterwards.
3. Log the exact error type.
4. Redo it later without notes.

Active recall beats passive rereading.

Objective reporting

Data analysis must remain neutral. Report both good and bad results, and choose summary measures that fairly represent the data.

- If data is skewed, do not report only the mean; include the median too.
- Do not handpick results to distort facts.
- Avoid emotional language like “disastrous crisis”; report the actual trend.
- Correlation does not equal causation: a hidden third variable may explain both patterns.

Big Picture: Numerical Descriptive Measures

The purpose of Chapter 3 and how all measures fit together.

Chapter 3 Big Picture

Chapter 3 answers: **How do we describe a dataset using numbers?**

Area	Question	Measures
Centre / location	Where is the data centred?	Mean, median, mode, geometric mean
Variation / spread	How spread out is it?	Range, variance, SD, CV, IQR
Relative position	Where does one value sit?	z-score, quartiles
Shape	What does the distribution look like?	Skewness, boxplot, outliers
Association	How do two variables move together?	Covariance, correlation

The Chapter 3 Golden Question

Before calculating, ask: **What is the question trying to describe?**

Most Chapter 3 mistakes happen because the wrong descriptive measure is chosen before the calculation begins.

Formula Sheet Decoder

Every Chapter 3 formula decoded into what it means, when to use it, and the biggest trap.

Sample Mean

$$\bar{x} = \Sigma x_i / n$$

The mean is the average or balance point of the data. Add all values, then divide by the number of observations.

Symbol	Meaning
$\bar{x}$	sample mean
x_i	each observed data value
n	number of observations
Σx_i	sum of all observations

Trap check

Trap	Fix
Forgetting to divide by n	Mean = total ÷ number of values
Outliers pull the mean	Use median if the data is skewed/extreme
Rounding too early	Keep extra decimals until the final answer

Sample Variance

$$s^2 = [\Sigma(x_i - \bar{x})^2] / (n - 1)$$

$$s^2 = \{ \Sigma x_i^2 - [(\Sigma x_i)^2 / n] \} / (n - 1)$$

Variance measures spread around the mean using squared distances. The shortcut formula is useful when sums are given.

Trap check

Trap	Fix
Using n instead of n−1	Sample variance uses n−1
Giving variance when SD is asked	Take $\sqrt{s^2}$
Forgetting squared deviations	Variance uses squared distances

Standard Deviation

$$s = \sqrt{s^2}$$

Standard deviation is the square root of variance and is in the original units of the data.

Trap check

Trap	Fix
Stopping at s^2	Take the square root
Taking the square root of the mean	Take the square root of the variance
Confusing s and s^2	s = SD, s^2 = variance

Coefficient of Variation

$$CV = (s / \bar{x}) \times 100\%$$

CV measures relative variation. It compares the standard deviation with the mean.

Trap check

Trap	Fix
Using variance in CV	CV uses s, not s^2
Forgetting ×100%	CV is usually a percentage
Comparing SDs with very different means	Use CV for relative spread

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

The simplest measure of spread. It only uses the smallest and largest values.

Trap check

Trap	Fix
Using $Q_3 - Q_1$	That is IQR
Thinking range uses all values	It only uses extremes
Ignoring outliers	Range is very outlier-sensitive

Median

```
Median position = (n + 1) / 2
```

The median is the middle value once the data is ordered. For even n, average the two middle values.

Trap check

Trap	Fix
Not ordering data	Always order first
Even n: choosing one middle value	Average the two middle values
Thinking outliers strongly affect median	Median is resistant

Mode

The mode is the most frequent value. There may be no mode, one mode, or multiple modes.

Type	Meaning
No mode	no repeated value
Unimodal	one mode
Bimodal	two modes
Multimodal	more than two modes

Trap check

Trap	Fix
Choosing the largest value	Mode = most frequent
Assuming mode always exists	It may not
Ignoring ties	There can be multiple modes

Geometric Mean

$$GM = \sqrt[n]{x_1 x_2 \ldots x_n}$$

Use geometric mean for multiplicative data, especially growth rates and returns. For returns, convert percentages to growth factors first.

Trap check

Trap	Fix
Using arithmetic mean for growth rates	Use geometric mean
Multiplying percentages directly	Convert to growth factors
Using zero/negative values casually	Basic GM needs positive values

Z-Score

$$z = (x - \bar{x}) / s$$

A z-score tells how many standard deviations a value lies from the mean.

z-score	Meaning
$z > 0$	above mean
$z < 0$	below mean
$z = 0$	equal to mean
$\|z\|$ large	more unusual

Trap check

Trap	Fix
Dividing by variance	Divide by SD
Ignoring sign	Positive/negative matters
Using $x + \bar{x}$	Use $x - \bar{x}$

Quartiles + IQR + Five-Number Summary

$IQR = Q_3 - Q_1$

Five-number summary = Minimum, Q_1, Median, Q_3, Maximum

Quartiles split ordered data into four parts. Q_2 is the median. IQR measures the middle 50% spread.

Trap check

Trap	Fix
Not ordering data	Order before quartiles
Using range as IQR	$IQR = Q_3 - Q_1$
Including mean in five-number summary	Use median, not mean

Grouped Data Mean + Variance

$\bar{x}\_f \approx \Sigma m_j f_j / n$

$s^2\_f \approx \{ \Sigma m_j^2 f_j - [(\Sigma m_j f_j)^2 / n] \} / (n - 1)$

Grouped data uses class midpoints because raw values are hidden. Results are approximate.

Trap check

Trap	Fix
Using class endpoints	Use midpoints
Forgetting frequency	Weight each midpoint by f
Giving grouped variance when SD is asked	Take the square root

Covariance + Correlation

$s\_xy = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$

$r = s\_xy / (s\_x\ s\_y)$

Covariance gives direction. Correlation gives strength + direction of a linear relationship and must be between −1 and 1.

Trap check

Trap	Fix
Sorting x and y separately	Keep pairs together
Using variances in r denominator	Use SDs
Saying correlation proves causation	It does not

Measure Recognition Table

A fast “which measure do I use?” guide for MCQs.

Main Measure Recognition Table

Question clue	Use	Why
average	Mean	arithmetic centre
middle value	Median	position-based centre
most frequent	Mode	frequency-based centre
growth rate / return	Geometric mean	multiplicative average
largest minus smallest	Range	simplest spread
spread around mean	Variance / SD	variation
standard deviation	$s = \sqrt{s^2}$	original units
relative variation	CV	$s/\bar{x} \times 100\%$
how many SDs from mean	Z-score	relative position
25th / 50th / 75th percentile	Quartiles	relative location
middle 50% spread	IQR	$Q_3 - Q_1$
boxplot	Five-number summary	min, Q1, median, Q3, max
class intervals + frequencies	Grouped data	use midpoints
move together	Covariance	direction
strength + direction	Correlation	standardised relationship
adding/multiplying values	Transformations	shift or scale rules

13-Step Decision Routine

1. Is it centre/location? Mean, median, mode.
2. Are there outliers/skewness? Median may be better.
3. Most common? Mode.
4. Growth/returns? Geometric mean.
5. Spread? Range, variance, SD, IQR.
6. Relative spread? CV.
7. Relative to mean? z-score.
8. 25%, 50%, 75%? Quartiles.
9. Boxplot? Five-number summary.
10. Grouped/class intervals? Midpoints.
11. Two variables move together? Covariance/correlation.
12. Strength + direction? Correlation.
13. Adding or multiplying all values? Transformation rules.

Central Tendency Question Cards

Mean, median, mode, geometric mean, and choosing the best centre.

Mean

$$\bar{x} = \Sigma x_i / n$$

Teach it simply: The mean is the ordinary average and the balance point of the data.

Example: For 4, 6, 8, 10: $\bar{x} = 28/4 = 7$.

Interpretation: The average value is 7.

Deeper explanation

What the mean is really doing

The mean spreads the total equally across all observations. For `4, 6, 8, 10`, the total is `28`. If that total is shared equally across 4 values, each value would be `7`.

$$\bar{x} = (4 + 6 + 8 + 10) / 4 = 7$$

Why the mean is called the balance point

The values below 7 are `4` and `6`. They are 3 and 1 units below the mean. The values above 7 are `8` and `10`. They are 1 and 3 units above the mean. The distances balance.

Value	Distance from mean 7
4	-3
6	-1
8	+1
10	+3

Why outliers matter

If the data were `4, 6, 8, 100`, the mean would jump to `29.5`, even though most values are much smaller. This is why the median is often safer when there is an extreme value.

Mean = fair share of the total, but it can be pulled by extreme values.

Trap check

Trap	Fix
Forgetting to divide by n	Mean = total ÷ number of values
Outliers	Median may be better

Median

```
Median position = (n + 1) / 2
```

Teach it simply: The median is the middle value after ordering.

Odd n example: 10, 4, 8, 6, 20 → order as 4, 6, 8, 10, 20. Median = 8.

Even n example: 4, 6, 8, 10 → median = (6+8)/2 = 7.

Deeper explanation

Why ordering is non-negotiable

The median is about position, not arithmetic. You cannot know the middle value until the values are in order.

Original order	Ordered data	Median
10, 4, 8, 6, 20	4, 6, 8, 10, 20	8

Odd vs even number of values

If there is an odd number of values, there is one middle value. If there is an even number, the middle is between two values, so you average them.

Data	Middle logic	Median
4, 6, 8, 10, 20	3rd value	8
4, 6, 8, 10	average 6 and 8	7

Why median survives outliers

In `4, 5, 5, 6, 100`, the median is still `5`. The huge value affects the mean badly, but it does not move the middle position much.

> **Median = middle after ordering. If there are outliers, trust median more than mean.**

Trap check

Trap	Fix
Not ordering data	Always order first
Even n	Average the two middle values

Mode

Teach it simply: The mode is the value that appears most often.

Example: 2, 3, 3, 4, 5, 5, 5, 6 → mode = 5.

Type	Meaning
No mode	no repeated value
Unimodal	one mode
Bimodal	two modes
Multimodal	more than two modes

Geometric Mean

$$GM = \sqrt[n]{x_1 x_2 \dots x_n}$$

Use it for: growth rates, returns, compound averages, multiplicative change.

Example: GM of 2, 4, 8 = $\sqrt[3]{64}$ = 4.

For returns, convert percentages to growth factors first: 10% increase → 1.10; 5% decrease → 0.95.

Deeper explanation

Why geometric mean is different

The ordinary mean adds values. The geometric mean multiplies values. That makes it useful when values compound, such as growth rates or investment returns.

$$GM = \sqrt[n]{x_1 x_2 ... x_n}$$

Example with plain numbers

For `2, 4, 8`, multiply first, then take the cube root because there are 3 values.

$$GM = \sqrt[3]{2 \times 4 \times 8} = \sqrt[3]{64} = 4$$

Growth-rate version

Returns must be converted to growth factors first. A 10% increase becomes `1.10`, and a 5% decrease becomes `0.95`. You do not multiply `10` and `-5` as percentages.

Return wording	Growth factor
10% increase	1.10
5% decrease	0.95
20% increase	1.20

Exam trap: arithmetic mean is for additive averages; geometric mean is for compound/multiplicative change.

Geometric mean = the average multiplier.

Choosing the Best Centre

Situation	Best measure
Symmetric data, no outliers	Mean
Skewed data or outliers	Median
Categorical/frequency question	Mode
Growth/returns	Geometric mean

Example: In 4, 5, 5, 6, 100, the mean is pulled upward, so the median is better.

Deeper explanation

The choice depends on the data shape

The “best” centre is not always the mean. You choose based on whether the data is balanced, skewed, or contains outliers.

Dataset type	Better centre	Reason
Symmetric, no extreme values	Mean	All values contribute fairly
Skewed or has outliers	Median	Position-based and resistant
Most common category/value	Mode	Frequency-based
Growth rates/returns	Geometric mean	Compounding is multiplicative

Quick comparison

For `4, 5, 5, 6, 7`, mean and median are both sensible. For `4, 5, 5, 6, 100`, the value 100 pulls the mean upward, so the median tells the typical centre better.

Outlier or skewness? Think median first.

Variation Question Cards

Range, variance, standard deviation, CV, and comparing spread.

Range

```
Range = Maximum − Minimum
```

For 4, 6, 8, 10: range = 10 − 4 = 6.

Range is simple but heavily affected by outliers.

Sample Variance

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

For 4, 6, 8, 10, $\bar{x} = 7$. Squared deviations are 9, 1, 1, 9, so the sum is 20. Sample variance = 20/3 = 6.6667.

Deeper explanation

What variance is measuring

Variance measures how far values are from the mean, but it squares the distances so that negative and positive deviations do not cancel out.

Value x	$x - \bar{x}$	$(x - \bar{x})^2$
4	-3	9
6	-1	1
8	1	1
10	3	9

The squared deviations add to:

```
9 + 1 + 1 + 9 = 20
```

Why divide by n−1?

Because this is **sample** variance. The sample mean is only an estimate of the true population mean, so using `n−1` adjusts for that estimation.

$$s^2 = 20 / (4 - 1) = 20 / 3 = 6.6667$$

What does 6.6667 mean?

It is the average squared spread around the mean. Because it is squared, it is harder to interpret directly than standard deviation.

Variance = squared spread. Sample variance divides by n−1.

Trap check

Trap	Fix
Using n	Sample variance uses n−1
Stopping at sum of squares	Divide by n−1

Standard Deviation

$$s = \sqrt{s^2}$$

From $s^2 = 6.6667$, $s = \sqrt{6.6667} \approx 2.5820$.

SD is in original units; variance is in squared units.

Deeper explanation

Why we square-root variance

Variance is in squared units. If the data is in minutes, variance is in minutes². Standard deviation takes the square root so the spread is back in the original units.

$$s = \sqrt{6.6667} \approx 2.5820$$

What does the answer mean?

The values typically vary by about **2.58 units** from the mean. It is not exact for each value; it is a general measure of spread.

Measure	Value	Unit logic
Variance	6.6667	squared units
Standard deviation	2.5820	original units

> **Exam trap:** if options include both `6.6667` and `2.5820`, read whether the question asks for variance or standard deviation.

> **Standard deviation = square root of variance = original-unit spread.**

Coefficient of Variation

$$CV = (s / \bar{x}) \times 100\%$$

If $\bar{x} = 7$ and $s = 2.5820$, then CV = 36.89%.

Use CV to compare relative spread between datasets with different means or units.

Deeper explanation

What CV is really comparing

CV compares the standard deviation to the mean. This makes spread relative instead of absolute.

```
CV = (s / x̄) × 100%
```

For $\bar{x} = 7$ and $s = 2.5820$:

```
CV = (2.5820 / 7) × 100% = 36.89%
```

What does 36.89% mean?

The standard deviation is about **36.89% of the mean**. That is why CV is useful when two datasets have different means.

Dataset	Mean	SD	CV	Interpretation
A	50	5	10%	more relative spread
B	200	10	5%	less relative spread

Dataset B has a larger SD, but Dataset A has greater relative variation because its CV is higher.

CV = relative spread. Use SD, not variance.

Comparing Spread

Dataset	Mean	SD	CV
A	50	5	10%
B	200	10	5%

Dataset B has larger SD, but Dataset A has larger relative variation because its CV is higher.

Z-Scores + Quartiles + Boxplots

Relative position, quartiles, IQR, boxplots and outlier checks.

Z-Score

```
z = (x − x̄) / s
```

If $x = 12$, $\bar{x} = 10$, $s = 2$, then $z = 1$. The value is 1 SD above the mean.

z	Meaning
positive	above mean
negative	below mean
0	equal to mean
large |z|	more unusual

Deeper explanation

What standardising does

A z-score converts a raw value into a standardised position. It tells you how far the value is from the mean in standard deviation units.

```
z = (x − x̄) / s
```

Example

If $x = 12$, $\bar{x} = 10$, and $s = 2$:

```
z = (12 − 10) / 2 = 1
```

This means 12 is **1 standard deviation above** the mean.

How to read the sign

z value	Meaning
positive	above the mean
negative	below the mean
zero	exactly at the mean

Exam trap: divide by standard deviation, not variance.

z-score = distance from mean measured in SDs.

Z-Score Outlier Check

```
|z| > 3 may indicate an outlier
```

If $x = 42$, $\bar{x} = 30$, $s = 4$, then $z = 3$. This is very unusual.

Large negative z-values can also be unusual.

Quartiles

Quartiles split ordered data into four parts.

Quartile	Meaning
Q_1	lower quartile / 25th percentile
Q_2	median / 50th percentile
Q_3	upper quartile / 75th percentile

For 3, 5, 7, 8, 10, 12, 13: $Q_1 = 5$, $Q_2 = 8$, $Q_3 = 12$.

Deeper explanation

Why the data must be ordered

Quartiles are position measures. They split an ordered dataset into four parts, so the first step is always to arrange the data from smallest to largest.

Example

For ordered data `3, 5, 7, 8, 10, 12, 13`:

Part	Values	Result
Median / Q_2	middle value	8
Lower half	3, 5, 7	$Q_1 = 5$
Upper half	10, 12, 13	$Q_3 = 12$

What the quartiles mean

$Q_1 = 5$ means roughly 25% of the data is at or below that lower quartile point. $Q_3 = 12$ marks the upper quartile point.

Method note: Different textbooks/software sometimes use slightly different quartile conventions. For this hub's worked examples, the median-of-halves method is used. If your tutorial memo uses a specific position method, follow that method in assessments.

Quartiles = ordered positions. Q_2 is the median.

Interquartile Range

$$IQR = Q_3 - Q_1$$

If $Q_1 = 5$ and $Q_3 = 12$, then IQR = 7. The middle 50% spans 7 units.

Deeper explanation

What IQR measures

The IQR measures the spread of the middle 50% of the data. It ignores the lowest 25% and highest 25%, so it is more resistant to outliers than the range.

$$IQR = Q_3 - Q_1$$

If $Q_1 = 5$ and $Q_3 = 12$:

$$IQR = 12 - 5 = 7$$

How to interpret it

The middle half of the data spans 7 units. This tells us about the central spread, not the total spread.

Spread measure	Uses	Outlier sensitivity
Range	minimum and maximum	high
IQR	Q_1 and Q_3	lower

IQR = middle 50% spread.

Five-Number Summary

$$\text{Minimum}, Q_1, \text{Median}, Q_3, \text{Maximum}$$

For 3, 5, 7, 8, 10, 12, 13, the five-number summary is 3, 5, 8, 12, 13.

Deeper explanation

Why this matters

The five-number summary is the numerical skeleton of a boxplot. Once you know these five values, you can draw and interpret the boxplot.

```
Minimum, Q₁, Median, Q₃, Maximum
```

Example

For `3, 5, 7, 8, 10, 12, 13`:

Value type	Value
Minimum	3
Q_1	5
Median	8
Q_3	12
Maximum	13

Common mistake

The mean is not part of the five-number summary. This summary is built around the median and quartiles.

Five-number summary = min, Q_1, median, Q_3, max. No mean.

Boxplot Interpretation

Boxplot part	Meaning
Box length	IQR
Line in box	Median
Long right whisker	Possible right skew
Long left whisker	Possible left skew
Median near centre	Roughly symmetric

Box length is IQR, not range.

Deeper explanation

What the boxplot shows

A boxplot is a picture of the five-number summary. The box itself runs from Q_1 to Q_3, so the box length is the IQR.

Feature	Meaning
Box length	middle 50% spread / IQR
Line inside box	median
Whiskers	spread outside the middle 50%
Far separate points	possible outliers

How to read skewness

Skewness follows the tail. A long right whisker suggests right skew. A long left whisker suggests left skew.

Exam trap: do not call the line in the box the mean. In a standard boxplot, it is the median.

Boxplot = five-number summary picture. Box = IQR.

IQR Outlier Rule

```
Lower fence = Q₁ − 1.5(IQR)
```

```
Upper fence = Q₃ + 1.5(IQR)
```

If $Q_1 = 5$, $Q_3 = 12$, IQR = 7, then lower fence = −5.5 and upper fence = 22.5.

Deeper explanation

What the fences do

The IQR outlier rule creates a lower and upper boundary. Values outside those boundaries are potential outliers.

```
Lower fence = Q₁ − 1.5(IQR)
```

```
Upper fence = Q₃ + 1.5(IQR)
```

Example

If `Q₁ = 5`, `Q₃ = 12`, and `IQR = 7`:

Fence	Calculation	Result
Lower	5 – 1.5(7)	-5.5
Upper	12 + 1.5(7)	22.5

Any value below -5.5 or above 22.5 would be flagged as a potential outlier.

Trap: the fence uses IQR, not range.

Outlier fences = Q_1 – 1.5IQR and Q_3 + 1.5IQR.

Grouped Data Question Cards

Midpoints, frequency, cumulative frequency, grouped mean and grouped SD.

Class Midpoint

```
m = (lower class limit + upper class limit) / 2
```

For class 10–20, midpoint = 15. Grouped data calculations use midpoints because raw values are hidden.

Deeper explanation

Why midpoints matter

In grouped data, the raw data values are hidden inside intervals. The midpoint is used as the best single representative for each class.

```
m = (lower limit + upper limit) / 2
```

For the class `10–20`:

```
m = (10 + 20) / 2 = 15
```

What we are pretending

If 5 values are in the class `10–20`, we do not know whether they were 11, 12, 18, etc. So for calculation purposes we estimate all 5 values as roughly 15.

Grouped data hides raw values, so midpoint becomes the representative value.

Relative Frequency

```
Relative frequency = f / n
```

For frequencies 2, 5, 3, total n = 10. Relative frequency for 10–20 is 5/10 = 0.50.

Deeper explanation

Frequency vs relative frequency

Frequency is the count. Relative frequency is the fraction or proportion of the total.

```
Relative frequency = f / n
```

For frequencies `2, 5, 3`, the total is:

```
n = 2 + 5 + 3 = 10
```

For the class with frequency 5:

```
5 / 10 = 0.50
```

Interpretation

A relative frequency of 0.50 means 50% of the observations are in that class.

Relative frequency = class count divided by total count.

Cumulative Relative Frequency

Cumulative relative frequency is a running total of relative frequencies.

Class	Frequency	Relative frequency	Cumulative relative frequency
0–10	2	0.20	0.20
10–20	5	0.50	0.70
20–30	3	0.30	1.00

Deeper explanation

What cumulative means

Cumulative means “adding as you go”. Each row includes that class plus all previous classes.

Class	Relative frequency	Cumulative relative frequency
0–10	0.20	0.20
10–20	0.50	0.20 + 0.50 = 0.70
20–30	0.30	1.00

Interpretation

By the end of the `10–20` class, 70% of observations are at or below the upper boundary of that class.

Check: the final cumulative relative frequency should be 1.00 or 100%.

Cumulative relative frequency = running total of proportions.

Median Interval

Use cumulative frequency. If n = 10, median position is around n/2 = 5. With cumulative frequencies 2, 7, 10, the 5th observation is in 10–20.

Deeper explanation

Why we get an interval, not an exact value

With grouped data, we do not know the exact raw values. So we identify the class interval where the median observation falls.

Step-by-step method

1. Find total frequency `n`.
2. Find the median position, often around `n/2`.
3. Use cumulative frequency.
4. The median interval is the first class where cumulative frequency reaches or passes the median position.

Class	Frequency	Cumulative frequency
0–10	2	2
10–20	5	7
20–30	3	10

Here `n = 10`, so the median position is around `5`. The first cumulative frequency to reach/pass 5 is 7, so the median interval is `10–20`.

Median interval = class where the middle observation falls.

Quartile Class

```
Q₁ position ≈ n/4
```

```
Q₃ position ≈ 3n/4
```

With n = 10, Q_1 position ≈ 2.5 and Q_3 position ≈ 7.5. Use cumulative frequency to find the class.

Deeper explanation

Same idea as median interval

Quartile classes use cumulative frequency positions. You are locating the class that contains Q_1 or Q_3.

```
Q₁ position ≈ n/4
```

```
Q₃ position ≈ 3n/4
```

Example with n = 10

Q_1 is around position `10/4 = 2.5`. Q_3 is around position `3(10)/4 = 7.5`.

Class	Cumulative frequency
0–10	2
10–20	7
20–30	10

The first cumulative frequency to pass 2.5 is 7, so Q_1 is in `10–20`. The first cumulative frequency to pass 7.5 is 10, so Q_3 is in `20–30`.

Quartile class = use cumulative frequency to find where the quartile position lands.

Approximate Grouped Mean

```
x̄_f ≈ Σmf / n
```

Using midpoints 5, 15, 25 and frequencies 2, 5, 3: Σmf = 160, n = 10, so x̄_f ≈ 16.

Deeper explanation

What the formula means

$\bar{x}_f \approx \Sigma mf / n$

Symbol	Meaning
$\bar{x}_f$	approximate grouped mean
m	midpoint of each class
f	frequency of each class
mf	midpoint × frequency
Σmf	total of all midpoint-frequency products
n	total number of observations

Why we use midpoints

For a class like `0–10` with frequency 2, we only know that 2 values were somewhere between 0 and 10. We do not know the exact raw values, so we estimate them using the midpoint 5.

Worked example

Class	Frequency f	Midpoint m	mf
0–10	2	5	10
10–20	5	15	75
20–30	3	25	75

$\Sigma mf = 10 + 75 + 75 = 160$

$n = 2 + 5 + 3 = 10$

$\bar{x}_f \approx 160 / 10 = 16$

What the answer means

The approximate average value is 16. It is approximate because the real raw values are hidden inside the class intervals.

Trap: do not just average the midpoints `(5+15+25)/3 = 15`. That ignores frequencies. The midpoint 15 appears 5 times, so it must count 5 times.

Grouped mean = weighted average of midpoints.

Approximate Grouped Standard Deviation

$s^2_f \approx \{\Sigma m^2 f - [(\Sigma mf)^2/n]\} / (n-1)$

With $\Sigma mf = 160$, $\Sigma m^2 f = 3050$, $n = 10$: $s^2_f = 54.4444$, so $s_f \approx 7.3786$.

Deeper explanation

Big idea

This is the grouped-data version of standard deviation: how far the estimated values are from the grouped mean, on average.

$$s^2\_f \approx \{\Sigma m^2 f - [(\Sigma mf)^2/n]\} / (n-1)$$

This formula gives the **variance first**. Then you take the square root to get standard deviation.

Concept version using the mean 16

We pretend the grouped values are roughly `5, 5, 15, 15, 15, 15, 15, 25, 25, 25`. The approximate mean is 16.

Midpoint m	Frequency f	m – 16	(m – 16)²	(m – 16)²f
5	2	-11	121	242
15	5	-1	1	5
25	3	9	81	243

$$242 + 5 + 243 = 490$$

$$s^2\_f = 490 / (10 - 1) = 54.4444$$

$$s\_f = \sqrt{54.4444} \approx 7.3786$$

Shortcut formula version

$$s^2\_f \approx [3050 - (160^2/10)] / 9 = [3050 - 2560] / 9 = 490/9 = 54.4444$$

Interpretation

The approximate grouped standard deviation is about 7.38. Values typically vary about 7.38 units from the approximate mean of 16.

Grouped SD = grouped variance first, then square-root.

Trap check

Trap	Fix
Giving variance when SD asked	Take the square root
Using class endpoints	Use midpoints
Treating result as exact	Grouped results are approximate

Empirical Rule + Chebyshev Cards

Spread around the mean: when to use 68–95–99.7 vs Chebyshev.

Empirical Rule

Use for approximately bell-shaped data.

Interval	Approximate proportion
Within 1 SD	68%
Within 2 SDs	95%
Within 3 SDs	99.7%

If $\bar{x} = 100$ and $s = 15$, about 95% lie between 70 and 130.

Deeper explanation

When you are allowed to use it

The empirical rule is only for data that is approximately bell-shaped. It is not a rule for every distribution.

Distance from mean	Approximate proportion	Memory
within 1 SD	68%	small central band
within 2 SDs	95%	most values
within 3 SDs	99.7%	almost all values

Example

If $\bar{x} = 100$ and $s = 15$, then within 2 SDs is:

```
100 − 2(15) = 70
```

```
100 + 2(15) = 130
```

So about 95% of observations lie between 70 and 130.

Trap: use SD, not variance, to build the interval.

Empirical rule = 68, 95, 99.7 for bell-shaped data.

Chebyshev's Rule

```
1 − 1/k², where k > 1
```

Works for any distribution shape and gives a minimum proportion.

k	At least proportion
2	75%
3	88.89%
4	93.75%

Deeper explanation

What makes Chebyshev different

Chebyshev's rule works for any distribution shape. It gives a minimum proportion, not an exact proportion.

```
1 − 1/k², where k > 1
```

Example for k = 2

```
1 − 1/2² = 1 − 1/4 = 0.75
```

So at least 75% of values lie within 2 standard deviations of the mean.

Why "at least" matters

The true proportion could be more than 75%. Chebyshev only guarantees the minimum.

k	Calculation	Minimum proportion
2	1 − 1/4	75%
3	1 − 1/9	88.89%
4	1 − 1/16	93.75%

Chebyshev = any shape, at least, k > 1.

Empirical vs Chebyshev

Feature	Empirical Rule	Chebyshev
Shape needed	Bell-shaped	Any shape
Answer type	Approximate	Minimum / at least
2 SDs	About 95%	At least 75%

"Bell-shaped" points to empirical. "At least" or "any distribution" points to Chebyshev.

Deeper explanation

How to choose between them

Look for wording about the shape of the distribution.

Question wording	Use	Why
bell-shaped / approximately normal	Empirical rule	shape condition is met
any distribution	Chebyshev	works without shape assumption
at least / minimum proportion	Chebyshev	Chebyshev gives a lower bound

2 SD comparison

For 2 standard deviations, empirical rule gives about 95% if bell-shaped, while Chebyshev gives at least 75% for any shape.

Trap: do not use 95% just because the question says "within 2 SDs". Check whether it says bell-shaped.

Bell-shaped = empirical. Any shape/minimum = Chebyshev.

Covariance + Correlation Cards

Paired data, covariance direction, correlation strength and interpretation.

Paired Data Setup

Covariance and correlation require paired observations. Keep each x matched with its y.

Observation	x	y
1	1	2
2	2	4
3	3	5
4	4	4

Do not sort x and y separately.

Deeper explanation

Why pairs must stay together

Covariance and correlation compare how two variables move across the same observations. If you sort x and y separately, you destroy the real pairing.

Observation	x	y	Pair
1	1	2	(1,2)
2	2	4	(2,4)
3	3	5	(3,5)
4	4	4	(4,4)

What paired data means

Each row belongs to one unit/person/item. The x and y values in that row must be analysed together.

> **Never sort x and y separately for covariance/correlation.**

Covariance

```
s_xy = Σ(xᵢ − x̄)(yᵢ − ȳ) / (n − 1)
```

Covariance shows direction of movement: positive = move together, negative = move oppositely, near 0 = weak/no linear movement.

Example covariance from the paired data is s_xy = 1.1667, which is positive.

Deeper explanation

What covariance is looking for

Covariance checks whether x and y tend to move in the same direction or opposite directions.

Covariance sign	Meaning
Positive	x above its mean tends to pair with y above its mean, and x below with y below
Negative	x above its mean tends to pair with y below its mean
Near 0	little/no linear movement pattern

Why the product matters

For each pair, we multiply $(x_i - \bar{x})$ and $(y_i - \bar{y})$. If both deviations have the same sign, the product is positive. If they have opposite signs, the product is negative.

Important: covariance gives direction, but its size is hard to interpret because it depends on the units of x and y.

Covariance sign tells direction of movement.

Correlation Coefficient

```
r = s_xy / (s_x s_y)
```

If s_xy = 1.1667, s_x = 1.2910, s_y = 1.2583, then r ≈ 0.718.

This suggests a positive, moderately strong linear relationship.

Deeper explanation

Why correlation is easier to interpret than covariance

Correlation standardises covariance by dividing by both standard deviations. That removes units and forces the answer between -1 and 1.

```
r = s_xy / (s_xs_y)
```

Example

```
r = 1.1667 / [1.2910(1.2583)] ≈ 0.718
```

Because the value is positive, the relationship is positive. Because it is fairly close to 1, it suggests a moderately strong linear relationship.

r	Interpretation
close to +1	strong positive linear relationship
close to -1	strong negative linear relationship
close to 0	weak/no linear relationship

Sanity check: if r is less than -1 or greater than +1, something went wrong.

Correlation = direction + strength of a linear relationship.

Interpreting Correlation

r value	Interpretation
close to +1	strong positive linear relationship
close to −1	strong negative linear relationship
close to 0	weak/no linear relationship
+1 or −1	perfect linear relationship

Correlation does not prove causation.

Deeper explanation

Two things to say every time

When interpreting correlation, mention both **direction** and **strength**.

Feature	What to look at	Example
Direction	sign of r	positive or negative
Strength	distance from 0	weak, moderate, strong

Correlation is not causation

Even a strong correlation does not prove that x causes y. There could be a third factor, coincidence, or a reverse relationship.

Exam wording: say "linear relationship" or "association", not "causes".

Correlation describes association, not cause.

Transformation Effects Cards

What happens to mean, variance, SD, IQR and CV when values are changed.

Adding a Constant

```
Y = X + c
```

Measure	Effect
mean / median / mode	add c
range	unchanged
IQR	unchanged
variance	unchanged
SD	unchanged

Adding shifts the whole dataset but does not change spread.

Deeper explanation

What adding does visually

Adding a constant shifts every value by the same amount. The whole dataset moves left or right/up or down, but the distances between values stay the same.

```
Y = X + c
```

Example

If values are `4, 6, 8, 10` and you add 5, they become `9, 11, 13, 15`. The centre increases, but the spread is unchanged.

Measure	Effect of adding c	Why
Mean/median/mode	add c	all positions shift
Range/IQR	unchanged	distances stay the same
Variance/SD	unchanged	spread around centre stays the same

Adding shifts centre, not spread.

Multiplying by a Constant

```
Y = aX
```

Measure	Effect
mean / median	multiply by a
range / IQR / SD	multiply by |a|
variance	multiply by a²

If $\bar{x} = 10$, $s^2 = 4$, $s = 2$ and $Y = 3X$, then $\bar{x}_y = 30$, $s^2_y = 36$, $s_y = 6$.

Deeper explanation

What multiplying does visually

Multiplying stretches or shrinks the dataset. Distances from the centre also stretch or shrink.

```
Y = aX
```

Example

If $\bar{x} = 10$, $s^2 = 4$, $s = 2$, and $Y = 3X$:

Measure	Calculation	Result
Mean	3(10)	30
SD	|3|(2)	6
Variance	3²(4)	36

Why variance uses a²

Variance is based on squared distances. If distances are multiplied by 3, squared distances are multiplied by $3^2 = 9$.

Multiplying scales spread; variance squares the multiplier.

Effect on CV

```
CV = (s / x̄) × 100%
```

Adding a constant usually changes CV because mean changes but SD does not. Multiplying by a positive constant leaves CV unchanged because mean and SD scale together.

Deeper explanation

Why adding and multiplying behave differently

CV depends on both the standard deviation and the mean.

```
CV = (s / x̄) × 100%
```

Adding a constant

Adding changes the mean but not the SD, so CV usually changes.

Before adding	After adding 5
$\bar{x} = 10$, s = 2, CV = 20%	$\bar{x} = 15$, s = 2, CV = 13.33%

Multiplying by a positive constant

Multiplying changes the mean and SD by the same factor, so their ratio stays the same.

Before multiplying	After multiplying by 3
$\bar{x} = 10$, s = 2, CV = 20%	$\bar{x} = 30$, s = 6, CV = 20%

Adding usually changes CV. Multiplying by a positive constant keeps CV unchanged.

Mixed Recognition Drill

Practise choosing the measure before calculating.

Drill 1: A question asks for the “average battery life” of a dataset.

Answer: Mean.

Why: “Average” usually means arithmetic mean.

Drill 2: A dataset has one extreme value. The question asks for the best measure of centre.

Answer: Median.

Why: Median is resistant to outliers.

Drill 3: A question asks for the most common observed value.

Answer: Mode.

Why: Mode is frequency-based.

Drill 4: A question asks for the average annual growth rate over several years.

Answer: Geometric mean.

Why: Growth rates compound multiplicatively.

Drill 5: A question asks for the difference between largest and smallest value.

Answer: Range.

Formula: maximum − minimum.

Drill 6: A question gives $s^2 = 25$ and asks for standard deviation.

Answer: Take the square root.

```
s = √25 = 5
```

Drill 7: A question asks for a relative measure of spread.

Answer: Coefficient of variation.

```
CV = (s / x̄) × 100%
```

Drill 8: A question asks how many standard deviations a value lies from the mean.

Answer: z-score.

```
z = (x − x̄) / s
```

Drill 9: A question asks for the middle 50% spread.

Answer: Interquartile range.

```
IQR = Q₃ − Q₁
```

Drill 10: A question asks for values needed to draw a boxplot.

Answer: Five-number summary.

```
Minimum, Q₁, Median, Q₃,
Maximum
```

Drill 11: A table gives class intervals and frequencies, and asks for approximate mean.

Answer: Grouped mean.

Why: Use class midpoints and frequencies.

Drill 12: A question asks for approximate SD from grouped data.

Answer: Grouped variance first, then square root.

Why: Use midpoints and frequencies.

Drill 13: A question asks whether two variables move together.

Answer: Covariance.

Why: Covariance gives direction of joint movement.

Drill 14: A question asks for strength and direction of a linear relationship.

Answer: Correlation.

Why: Correlation standardises covariance.

Drill 15: Each value in a dataset is increased by 2. What happens to SD?

Answer: SD unchanged.

Why: Adding a constant shifts values but does not change spread.

Final Trap Sheet + One-Page Checklist

Use this immediately before the A2 or before timed MCQs.

One-Page Checklist

Question asks...	Use
average	mean
middle	median
most common	mode
growth / return	geometric mean
largest – smallest	range
spread around mean	variance / SD
relative spread	CV
how many SDs from mean	z-score
middle 50%	IQR
boxplot	five-number summary
class intervals	grouped midpoint methods
bell-shaped proportions	empirical rule
any-shape minimum proportion	Chebyshev
two variables move together	covariance
strength + direction	correlation
adding/multiplying values	transformation rules

Danger Words

Danger word / phrase	Pause because...
standard deviation	square root of variance
variance	do not square-root unless SD asked
relative variation	CV
middle 50%	IQR
approximate	grouped data likely
bell-shaped	empirical rule
at least	Chebyshev often
paired data	covariance/correlation
strength and direction	correlation
increased by	adding constant
multiplied by	scaling transformation

10-Second MCQ Routine

1. Is the question asking centre, spread, position, shape, or relationship?
2. Does data need to be ordered?
3. Are there outliers or skewness?
4. Variance or SD?
5. Relative spread?
6. Grouped data? Use midpoints.
7. Paired data? Keep pairs together.
8. Covariance or correlation?
9. Adding or multiplying transformation?
10. Does the answer make sense?

Final Memory Table

Topic	One-line rule
Mean	average / balance point
Median	middle after ordering
Mode	most frequent
Geometric mean	compound/growth average
Range	max – min
Variance	squared spread
SD	√variance
CV	relative spread %
Z-score	SDs from mean
IQR	$Q_3 - Q_1$
Grouped mean	midpoint × frequency
Empirical rule	68, 95, 99.7 for bell-shaped data
Chebyshev	$1 - 1/k^2$, any shape
Covariance	direction
Correlation	strength + direction
Adding constant	centre shifts, spread unchanged
Multiplying constant	centre + spread scale; variance squares

Mini MCQ Practice Set

40 MCQs with answer reveals, explanations, and persistent score tracking.

Q1

Not attempted

For the data: 4, 6, 8, 10, find the mean.

1. **A.** 6
2. **B.** 7
3. **C.** 8
4. **D.** 10

Answer: B

$\bar{x} = (4+6+8+10)/4 = 28/4 = 7$. The mean is the arithmetic average.

Q2

Not attempted

For the data: 10, 4, 8, 6, 20, find the median.

1. **A.** 6
2. **B.** 8
3. **C.** 10
4. **D.** 20

Answer: B

Order first: 4, 6, 8, 10, 20. There are 5 values, so the middle value is the 3rd value: 8.

Q3

Not attempted

For the data: 4, 6, 8, 10, find the median.

1. **A.** 6
2. **B.** 7
3. **C.** 8
4. **D.** 9

Answer: B

There are 4 values, so average the two middle values: (6+8)/2 = 7.

Q4

Not attempted

For the data: 2, 3, 3, 4, 5, 5, 5, 6, find the mode.

1. **A.** 3
2. **B.** 4
3. **C.** 5
4. **D.** 6

Answer: C

The mode is the most frequent value. The value 5 appears three times.

Q5

Not attempted

Which measure of centre is usually best when a dataset has a strong outlier?

1. **A.** Mean
2. **B.** Median
3. **C.** Range
4. **D.** Standard deviation

Answer: B

The median is resistant to outliers. The mean can be pulled upward or downward by extreme values.

Q6

Not attempted

Find the geometric mean of 2, 4, 8.

1. **A.** 3
2. **B.** 4
3. **C.** 4.67
4. **D.** 8

Answer: B

GM = $\sqrt[3]{(2\times4\times8)} = \sqrt[3]{64} = 4$. Geometric mean is useful for multiplicative data.

Q7

Not attempted

For the data: 4, 6, 8, 10, find the range.

1. **A.** 4
2. **B.** 5
3. **C.** 6
4. **D.** 14

Answer: C

Range = maximum − minimum = 10 − 4 = 6.

Q8

Not attempted

For the data: 4, 6, 8, 10, the mean is 7. What is the sum of squared deviations from the mean?

1. **A.** 10
2. **B.** 16
3. **C.** 20
4. **D.** 28

Answer: C

Squared deviations: $(-3)^2 + (-1)^2 + 1^2 + 3^2 = 9+1+1+9 = 20$.

Q9

Not attempted

Using the same data 4, 6, 8, 10, find the sample variance.

1. **A.** 5
2. **B.** 6.6667
3. **C.** 2.5820
4. **D.** 20

Answer: B

Sample variance uses n−1: $s^2 = 20/(4-1) = 20/3 = 6.6667$.

Q10

Not attempted

If $s^2 = 6.6667$, find the standard deviation.

1. **A.** 6.6667
2. **B.** 3.3334
3. **C.** 2.5820
4. **D.** 1.6667

Answer: C

$s = \sqrt{s^2} = \sqrt{6.6667} \approx 2.5820$. Do not stop at variance when SD is asked.

Q11

Not attempted

A dataset has $\bar{x} = 7$ and s = 2.5820. Find the coefficient of variation.

1. **A.** 2.5820%
2. **B.** 7%
3. **C.** 36.89%
4. **D.** 271.10%

Answer: C

CV = $(s/\bar{x})\times100\% = (2.5820/7)\times100\% = 36.89\%$.

Q12

Not attempted

Dataset A has mean 50 and SD 5. Dataset B has mean 200 and SD 10. Which dataset has greater relative variation?

1. **A.** Dataset A
2. **B.** Dataset B
3. **C.** They have equal relative variation
4. **D.** Cannot be determined

Answer: A

$CV_A = 5/50\times100\% = 10\%$. $CV_B = 10/200\times100\% = 5\%$. Dataset A has greater relative variation.

Q13 Not attempted

Suppose $x = 12$, $\bar{x} = 10$, and $s = 2$. Find the z-score.

1. **A.** –1
2. **B.** 0
3. **C.** 1
4. **D.** 2

Answer: C

$z = (x-\bar{x})/s = (12-10)/2 = 1$. The value is 1 SD above the mean.

Q14 Not attempted

Suppose $x = 42$, $\bar{x} = 30$, and $s = 4$. Find the z-score.

1. **A.** 2
2. **B.** 3
3. **C.** 4
4. **D.** 12

Answer: B

$z = (42-30)/4 = 3$. This is very unusual because it is 3 SDs above the mean.

Q15 Not attempted

For the ordered data 3, 5, 7, 8, 10, 12, 13, find Q_1.

1. **A.** 3
2. **B.** 5
3. **C.** 7
4. **D.** 8

Answer: B

Median is 8. Lower half is 3, 5, 7, so $Q_1 = 5$.

Q16 Not attempted

For the ordered data 3, 5, 7, 8, 10, 12, 13, find Q_3.

1. **A.** 10
2. **B.** 12
3. **C.** 13
4. **D.** 8

Answer: B

Upper half is 10, 12, 13, so $Q_3 = 12$.

Q17 Not attempted

If $Q_1 = 5$ and $Q_3 = 12$, find the IQR.

1. **A.** 5
2. **B.** 7
3. **C.** 12
4. **D.** 17

Answer: B

$IQR = Q_3 - Q_1 = 12 - 5 = 7$.

Q18 Not attempted

For the data 3, 5, 7, 8, 10, 12, 13, what is the five-number summary?

1. **A.** 3, 5, 8, 12, 13
2. **B.** 3, 7, 8, 10, 13
3. **C.** 5, 7, 8, 10, 12
4. **D.** 3, 5, 7, 12, 13

Answer: A

Five-number summary = minimum, Q_1, median, Q_3, maximum = 3, 5, 8, 12, 13.

Q19 Not attempted

If $Q_1 = 5$, $Q_3 = 12$, and $IQR = 7$, find the upper outlier fence.

1. **A.** 1.5
2. **B.** 10.5
3. **C.** 17.5
4. **D.** 22.5

Answer: D

Upper fence = $Q_3 + 1.5(IQR) = 12 + 1.5(7) = 22.5$.

Q20 Not attempted

A boxplot has a much longer right whisker than left whisker. What does this usually suggest?

1. **A.** Left skew
2. **B.** Right skew
3. **C.** Perfect symmetry
4. **D.** No variation

Answer: B

Skewness follows the tail. A long right whisker suggests right skew.

Q21

Not attempted

For grouped classes 0–10, 10–20, 20–30, what is the midpoint of the class 10–20?

1. **A.** 10
2. **B.** 15
3. **C.** 20
4. **D.** 30

Answer: B

Midpoint = (10+20)/2 = 15.

Q22

Not attempted

For frequencies 2, 5, 3, what is the relative frequency of class 10–20?

1. **A.** 0.20
2. **B.** 0.30
3. **C.** 0.50
4. **D.** 0.70

Answer: C

Total n = 2+5+3 = 10. Relative frequency = 5/10 = 0.50.

Q23

Not attempted

For frequencies 2, 5, 3, what is the cumulative relative frequency up to and including 10–20?

1. **A.** 0.20
2. **B.** 0.50
3. **C.** 0.70
4. **D.** 1.00

Answer: C

Cumulative relative frequency = 0.20 + 0.50 = 0.70.

Q24

Not attempted

For frequencies 2, 5, 3 across classes 0–10, 10–20, 20–30, which class is the median interval?

1. **A.** 0–10
2. **B.** 10–20
3. **C.** 20–30
4. **D.** Cannot be determined

Answer: B

n = 10, median position ≈ n/2 = 5. Cumulative frequencies are 2, 7, 10, so the 5th observation falls in 10–20.

Q25

Not attempted

Using midpoints for classes 0–10, 10–20, 20–30 with frequencies 2, 5, 3, find the approximate grouped mean.

1. **A.** 10
2. **B.** 15
3. **C.** 16
4. **D.** 20

Answer: C

Midpoints are 5, 15, 25. Σmf = 10+75+75 = 160 and n = 10, so x̄_f ≈ 16.

Q26

Not attempted

For grouped data, suppose Σmf = 160, $\Sigma m^2 f$ = 3050, and n = 10. Find the approximate grouped sample standard deviation.

1. **A.** 54.4444
2. **B.** 7.3786
3. **C.** 16
4. **D.** 490

Answer: B

s²_f ≈ [3050 − (160²/10)]/(10−1) = 490/9 = 54.4444. SD = √54.4444 = 7.3786.

Q27

Not attempted

The empirical rule applies mainly to which type of data?

1. **A.** Any distribution
2. **B.** Approximately bell-shaped data
3. **C.** Only categorical data
4. **D.** Only grouped data

Answer: B

The empirical rule applies to approximately bell-shaped data.

Q28

Not attempted

For approximately bell-shaped data, what proportion lies within 2 standard deviations of the mean?

1. **A.** 68%
2. **B.** 75%
3. **C.** 95%
4. **D.** 99.7%

Answer: C

Empirical rule: within 1 SD ≈ 68%, within 2 SDs ≈ 95%, within 3 SDs ≈ 99.7%.

Q29

Not attempted

A bell-shaped dataset has $\bar{x} = 100$ and $s = 15$. Approximately 95% of observations lie between:

1. **A.** 85 and 115
2. **B.** 70 and 130
3. **C.** 55 and 145
4. **D.** 100 and 130

Answer: B

For 95%, use ±2 SDs: $100 - 2(15) = 70$ and $100 + 2(15) = 130$.

Q30

Not attempted

Chebyshev's rule can be used for:

1. **A.** Only bell-shaped data
2. **B.** Only normal data
3. **C.** Any distribution shape
4. **D.** Only categorical data

Answer: C

Chebyshev's rule applies to any distribution shape.

Q31

Not attempted

Using Chebyshev's rule, what is the minimum proportion of observations within 2 SDs of the mean?

1. **A.** 50%
2. **B.** 68%
3. **C.** 75%
4. **D.** 95%

Answer: C

$1 - 1/k^2 = 1 - 1/2^2 = 1 - 1/4 = 0.75$. At least 75%.

Q32

Not attempted

Which statement is correct?

1. **A.** Empirical rule works for any distribution
2. **B.** Chebyshev gives exact percentages
3. **C.** Empirical rule gives approximate percentages for bell-shaped data
4. **D.** Chebyshev can only be used when data is normal

Answer: C

Empirical rule gives approximate percentages for bell-shaped data. Chebyshev gives minimum proportions for any shape.

Q33

Not attempted

What does a positive covariance suggest?

1. **A.** As x increases, y tends to decrease
2. **B.** As x increases, y tends to increase
3. **C.** The variables are unrelated in every possible way
4. **D.** x causes y

Answer: B

Positive covariance means the variables tend to move in the same direction.

Q34

Not attempted

Using sxy = 1.1667, sx = 1.2910, sy = 1.2583, find the correlation coefficient.

1. **A.** 0.718
2. **B.** 1.1667
3. **C.** 1.549
4. **D.** −0.718

Answer: A

$r = sxy/(sx\ sy) = 1.1667/[1.2910(1.2583)] \approx 0.718$.

Q35

Not attempted

How should $r \approx 0.718$ be interpreted?

1. **A.** Weak negative linear relationship
2. **B.** No linear relationship
3. **C.** Positive, moderately strong linear relationship
4. **D.** Perfect positive relationship

Answer: C

The value is positive and fairly close to 1, so it suggests a positive, moderately strong linear relationship.

Q36

Not attempted

Which statement about correlation is correct?

1. **A.** Correlation can be greater than 1
2. **B.** Correlation proves causation
3. **C.** Correlation measures strength and direction of a linear relationship
4. **D.** Correlation is always the same as covariance

Answer: C

Correlation measures strength and direction of a linear relationship. It must be between −1 and 1 and does not prove causation.

Q37

Not attempted

If every value in a dataset is increased by 5, what happens to the standard deviation?

1. **A.** It increases by 5
2. **B.** It decreases by 5
3. **C.** It stays the same
4. **D.** It is multiplied by 25

Answer: C

Adding a constant shifts the dataset but does not change spread.

Q38

Not attempted

Suppose $\bar{x} = 10$, $s^2 = 4$, and $s = 2$. If $Y = X + 5$, what is the variance of Y?

1. **A.** 4
2. **B.** 9
3. **C.** 14
4. **D.** 25

Answer: A

Adding a constant does not change variance, so s²_Y = 4.

Q39

Not attempted

Suppose $\bar{x} = 10$, $s^2 = 4$, and $s = 2$. If $Y = 3X$, what is the variance of Y?

1. **A.** 6
2. **B.** 12
3. **C.** 36
4. **D.** 81

Answer: C

When multiplying by a, variance is multiplied by a². s²_Y = 3²(4) = 36.

Q40

Not attempted

Suppose $\bar{x} = 10$ and $s = 2$. The CV is 20%. If every value is multiplied by 3, what happens to the CV?

1. **A.** It becomes 60%
2. **B.** It becomes 6.67%
3. **C.** It stays 20%
4. **D.** It becomes 23%

Answer: C

Multiplying all values by a positive constant multiplies both mean and SD by the same factor, so CV stays the same.

Error Log

Turn every wrong MCQ into a fixable pattern.

Error Category Codes

Code	Error category	Meaning
REC	Measure Recognition	wrong measure/method
ORD	Ordering Error	forgot to order data
CENT	Centre / Location	mean, median, mode, GM
GM	Geometric Mean	growth-factor mistake
VAR	Variance	variance calculation/interpretation
SD	Standard Deviation	forgot square root or confused s/s^2
CV	Coefficient of Variation	used variance, forgot % or relative spread
Z	Z-score	wrong formula/sign/variance used
QRT	Quartiles / IQR	mixed Q_1, Q_3, IQR, positions
BOX	Boxplot / Shape	misread five-number summary/skewness
OUT	Outlier Rule	wrong fence or z rule
GRP	Grouped Data	midpoints/frequencies/cumulative/approximation
EMP	Empirical Rule	68–95–99.7 or shape mistake
CHEB	Chebyshev	forgot formula, at least, or k>1
COV	Covariance	pairs/sign/denominator
CORR	Correlation	SD denominator, r range, causation
TRANS	Transformations	adding vs multiplying
CALC	Calculator	input/square-root/sum error
CARE	Careless Error	copying/rounding/reading
INT	Interpretation	answer not meaningful

Main error log

Q no.	Topic	My answer	Correct answer	Error code	What went wrong?	Correct method	Redo date	Fixed?
								☐
								☐
								☐
								☐
								☐
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Redo Rule

Attempt	Action
Attempt 1 wrong	Log the error
Same-day redo	Redo after reviewing correct method
Next-day redo	Redo without notes
Final redo	Mix with other Chapter 3 topics

Only tick “Fixed” when the next-day redo is correct without looking at the memo.

Trap Checklist

- [ ] Identified centre/spread/position/shape/relationship
- [ ] Ordered data for median/quartiles
- [ ] Took square root if SD was asked
- [ ] Used SD, not variance, for CV
- [ ] Used $z = (x - \bar{x})/s$
- [ ] Used $IQR = Q_3 - Q_1$
- [ ] Used midpoints for grouped data
- [ ] Remembered grouped answers are approximate
- [ ] Used empirical only for bell-shaped data
- [ ] Remembered Chebyshev gives “at least”
- [ ] Kept x,y pairs together
- [ ] Checked r is between −1 and 1
- [ ] Avoided correlation = causation
- [ ] Distinguished adding vs multiplying